UPC : 0013512801

Total Pages : 3

Roll No. ....

B.A./B.Sc. (Fifth Semester)

Examination, 2025-26

MATHEMATICS

Paper – I (Major)

[Abstract Algebra]

Time : 3 Hours] **[Maximum Marks : 75**

Note : The question paper is divided into *two* sections 'A' and 'B'. Attempt any *five* questions from Section 'A' and *three* from Section 'B'. Marks are indicated against each section.

SECTION–A

(Short Answer Type Questions)

Note : Attempt any *five* questions from the following *eight* questions. Each question carries 6 marks. (5×6=30)

1. Let H = {(1), (12)}. Is H normal in S_3?

[P.T.O.

2. Show that $U(8)$ is not isomorphic to $U(10)$ and isomorphic to $U(12)$.

3. How many homomorphisms are there from Z_{20} onto Z_8 and Z_{20} onto Z_4?

4. Find all automorphism of the group Z_{12}.

5. Find all the Sylow 3-subgroups of S_4.

6. Find all maximum ideals in :

(a) Z_8,

(b) Z_{10},

(c) Z_{12}.

7. Show that every subgroup of an abelian group is normal. 4/

8. Show that the set $S = \{1, 5, 7, 11\}$ is a group with respect to the multiplication modulo 12. CFull 4/

SECTION-B

(Long Answer Type Questions)

Note : Attempt any *three* questions from the following *five* questions. Each question carries 15 marks. (3×15=45)

9. Let

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 3 & 5 & 4 & 6 \end{bmatrix} \text{ and } \beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 2 & 4 & 3 & 5 \end{bmatrix}$$

Compute each of the following :

(a) α^{-1}.

(b) $\beta\alpha$. (Full)

(c) $\alpha\beta$. (Full)

10. Prove that a group G is abelian if and only if $(ab)^{-1} = a^{-1}b^{-1}$ for all a and b in G. (Full)

11. Prove that the following rings are integral domains :

(a) The ring of integers.

(b) The ring of Gaussian integer $Z[i] = \{a + bi \mid a, b \in Z\}$

(c) The ring $Z[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in Z\}$.

12. Prove that :

(a) Every field is an integral domain.

(b) Finite integral domain is field.

13. State and prove Lagrange's theorem. (Full)